

NiTiNOL Tyre Research analysis

-this doc presents research analysis.

Set of wires have been over-under woven

Slinky around the rim , polymer ke inside
slinky 100psi

Puncture with no loss of performance

Bulletproof -

We cant use rubber pneumatic tires bcs of
extreme condition on moon and mars
having low pressure there is no confining
pressure outside in surface which cost
exploration in it

Temp . dropping extreme lows rubber
become brittle

-90 is glass transition temperature where
polymer going being from flexible to rigid
like brittle

Rubber dipping in liquid nitrogen ,
lightweight , 0.7 mm thick skin , grabbing
soil wheels having more peak load s in
area given in grousers



there is big **holes and cracks at those areas**

Stress produce spacing in wheel

Kinked and stretched 0.3-0.8 % test

Apollo lvr panograph



This is on the surface here

Ripped and strengthen prolen is that
dosent sink to ground with maintaining
flexibility

It is to minimize plastic deformataion with
under the mess

Internal structure - bump stop = it keep it
just below the proportional limit of
deformation , to prevent deformtion ,
having no fracture

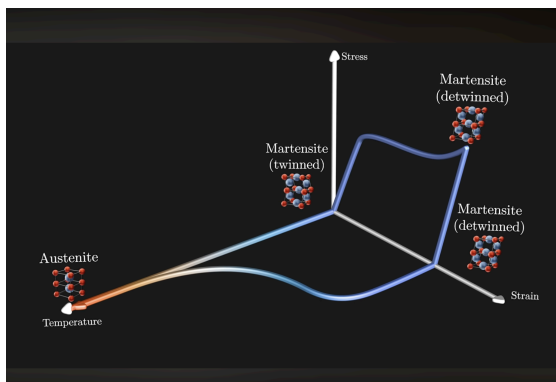
Repeatedly worked Heated , cooled
Pipe smoker - NiTiNOL

Nickel + titanium = alloy got
undergo phase change in solid
state

Heated austentite ,Twinned
martensite on when cooled ,
detwineed martensite changing
crystal structure stress applied it
goes to austentite on by heating

"Solid state phase transformation -
 - - releasing entropy - - - while wire
 transforming into round wire
 shaped it is exothermic process - -
 stress removed then phase change
 into austenite —it is endothermic "

SHAPE MEMORY ALLOY - can also
 generate significant forces when
 they're heated , which means they
 can also be used as actuators .



To fracture of rock — we use those
 forces

*extra—vortex generator

-150 -350degree centigrade

Main flaps on aircrafts controlling
 by heating element

For durable wheels

6-8% strain in wires , super or
 pseudo elasticity

NiTiNOL having transition

temperature below than room
 temperature

Austenite to dewtwinned

martensite - - - which gives us the
 strain prop and elasticity and
 elongness in it

Stress induced phase change



Field of elastocalorics - - - where
 heat is used as Heat pump

Dissipation in bending of straight
 wire into round is same as shock
 absorber in tyre

Then tyre has a complete potential
 of complete suspension system

- - - it also increases load
 bearing capacity with
 sharing load on all area
 around tyres not just on
 bottom or any subjective
 point
 - They weave nitinol spring
 together into a mesh (684
 times per tyre)
 - Terrain endurance - zero slip
 condn =6.7cm/s
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